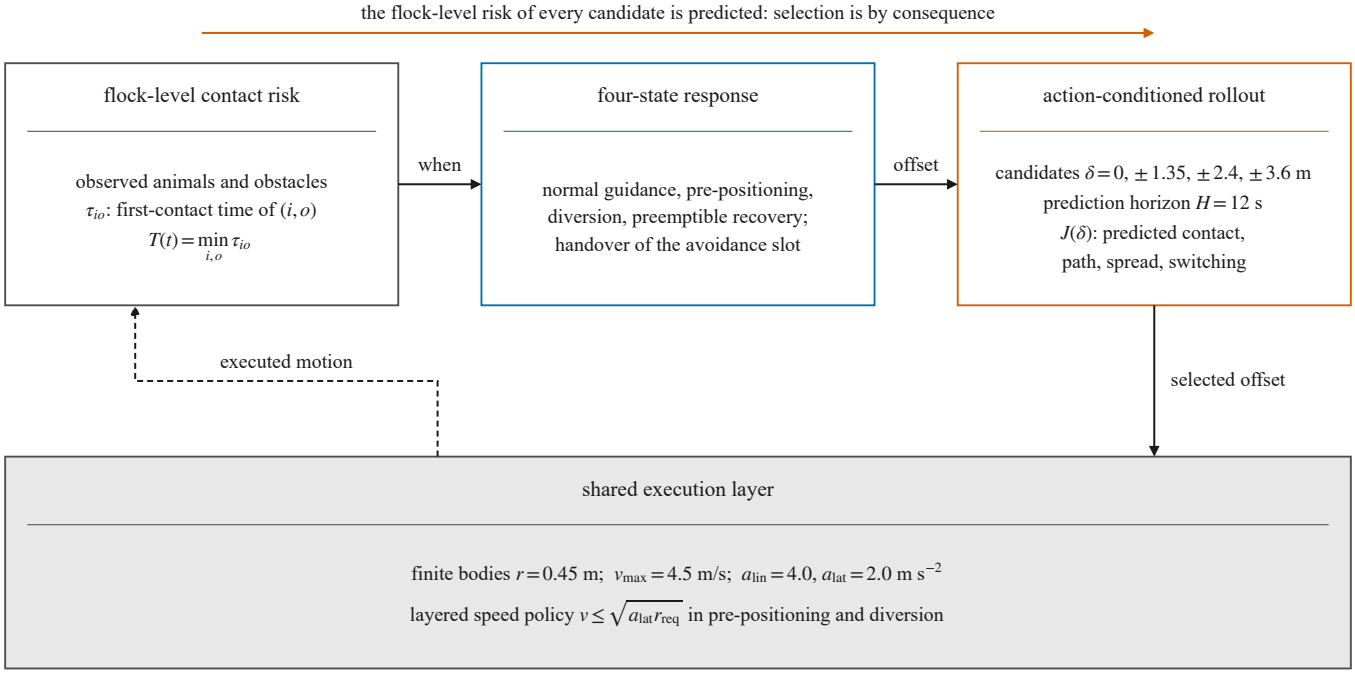
the flock-level risk of every candidate is predicted: selection is by consequence
flock-level contact risk
observed animals and obstacles
τ_{io}: first-contact time of (i, o)
$T(t) = \min_{i,o} \tau_{io}$
when
four-state response
normal guidance, pre-positioning,
diversion, preemptible recovery;
handover of the avoidance slot
offset
action-conditioned rollout
candidates $\delta = 0, \pm 1.35, \pm 2.4, \pm 3.6$ m
prediction horizon $H = 12$ s
$J(\delta)$: predicted contact,
path, spread, switching
executed motion
selected offset
shared execution layer
finite bodies $r = 0.45$ m; $v_{max} = 4.5$ m/s; $a_{lin} = 4.0$, $a_{lat} = 2.0$ m s^{-2}
layered speed policy $v \le \sqrt{a_{lat} r_{req}}$ in pre-positioning and diversion